LUNAR LANDER AI

Reinforcement Learning with PPO

Project report · Gymnasium LunarLander-v3 · Stable-Baselines3

Project objective

Train an AI agent to control a simulated lunar lander using reinforcement learning, then compare an initial policy with a more extensively trained checkpoint.

Executive summary

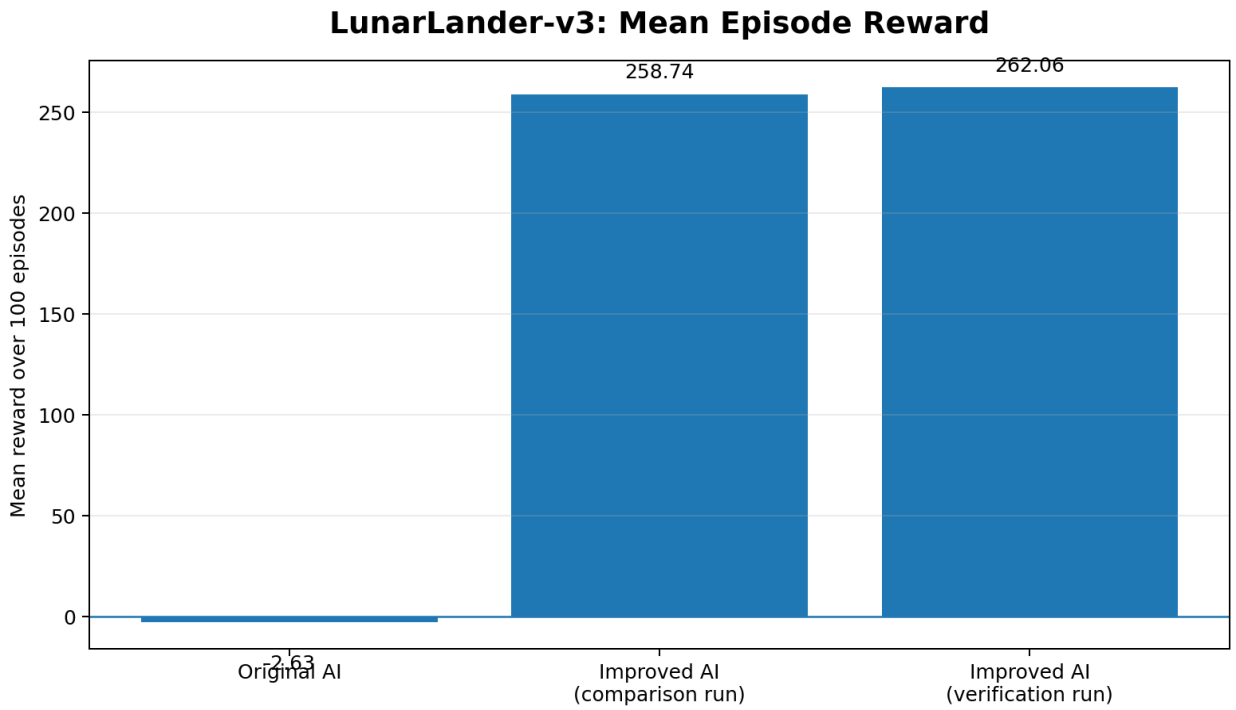
The original model recorded a mean reward of -2.63 over 100 episodes. The improved checkpoint recorded mean rewards of 258.74 in the comparison run and 262.06 in a subsequent verification run. These results indicate a large improvement in the task's reward metric. Landing counts below are heuristic final-state checks, not definitive success labels.

Evaluation run	Mean reward	Std. dev.	Landing-state indicators
Original AI	-2.63	37.92	8/100 (previous check)
Improved AI — comparison	258.74	46.34	85/100 (previous check)
Improved AI — verification	262.06	55.49	84/100 (stricter check)

1. Tools and approach

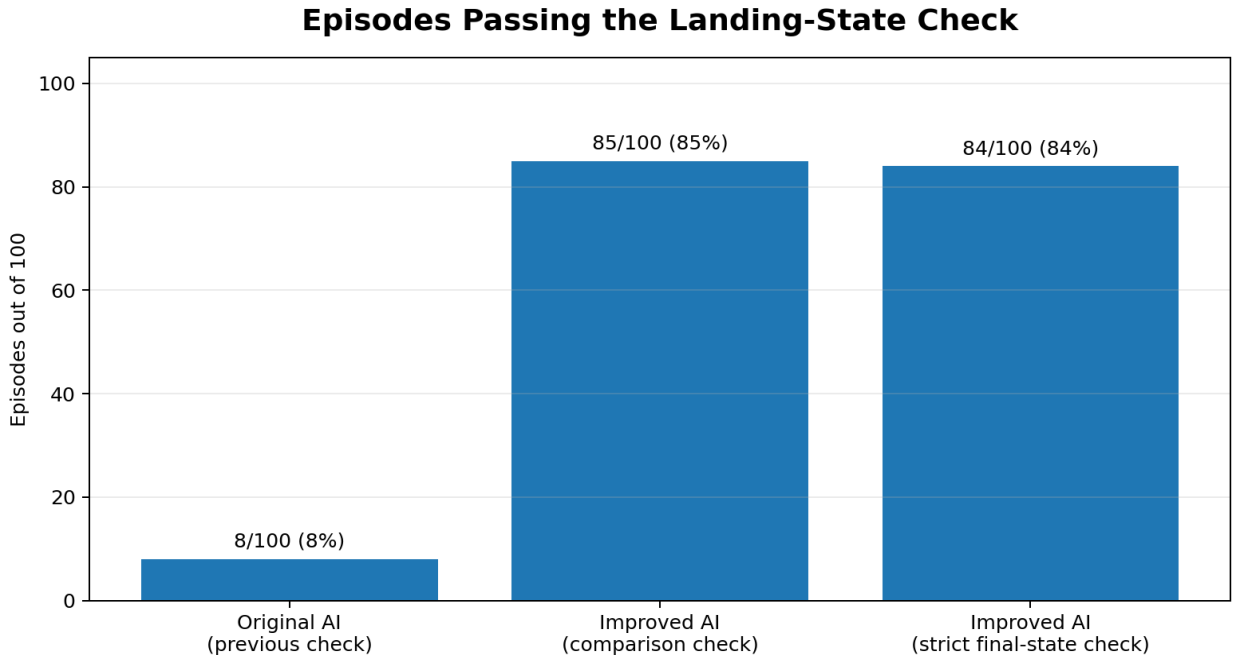
Gymnasium provides the LunarLander-v3 simulation. Stable-Baselines3 provides PPO (Proximal Policy Optimization), an on-policy reinforcement-learning algorithm. The agent observes the environment state and chooses actions to control the lander. The improved checkpoint was continued from the original model for approximately one million additional timesteps.

2. Reward results



The mean reward increased from -2.63 for the original model to above 258 in both reported improved-model runs. These are separate evaluation runs, so small differences between improved-model means are expected.

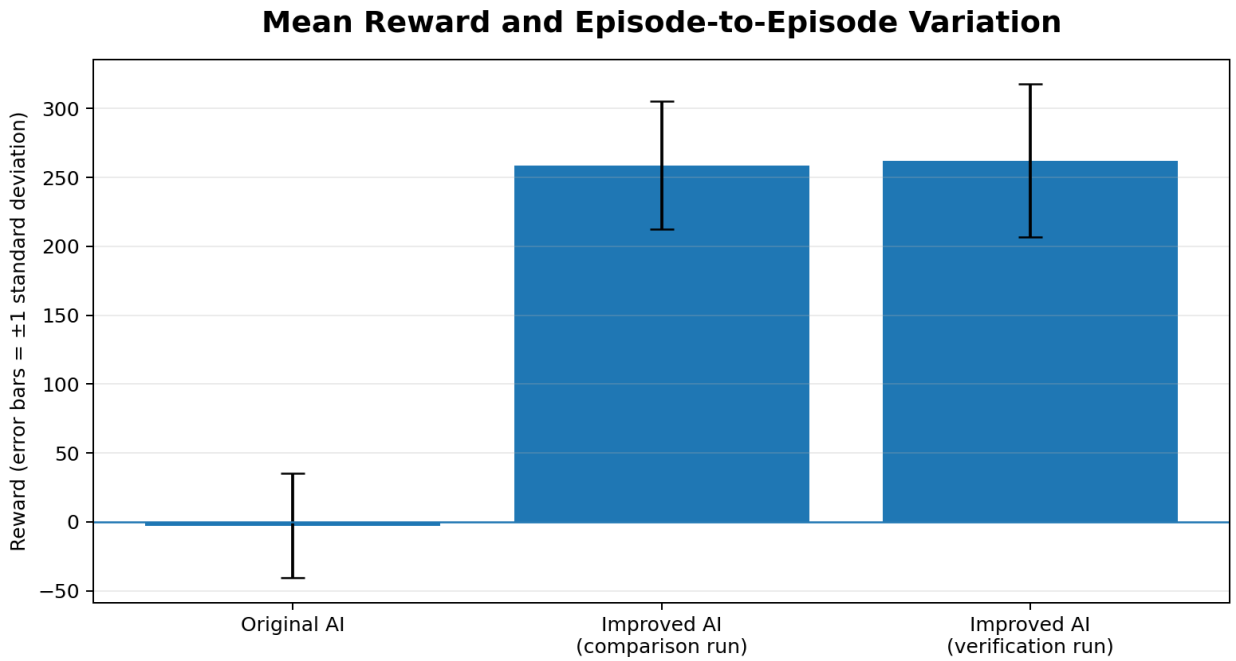
3. Landing-state indicators



Checks are heuristic indicators, not definitive environment success labels; criteria differed slightly.

The verification script reported 84 of 100 episodes satisfying its stricter final-state thresholds: both legs touching, near-upright angle, and low horizontal and vertical velocity. It also reported 13 terminated episodes that did not pass this check and 3 truncated episodes. Because a final observation is not a dedicated success label, these figures must be described as indicators, not confirmed landings.

4. Reward variability



Error bars show spread across episodes; they are not confidence intervals.

5. Reward hacking: educational demonstration

The pack includes a separate toy example in which a flawed reward gives points for any movement. An agent can earn points while moving away from the target. A better proxy rewards progress toward the target. This demonstrates the general risk of reward misspecification; it does not mean the trained LunarLander agent itself has been shown to reward-hack.

6. Limitations and next improvements

• Evaluation results vary with random episode seeds.
• The standard deviation measures episode-to-episode spread; it is not a confidence interval.
• The landing-state thresholds are heuristic and should not be treated as a definitive success rate.
• For a stronger scientific evaluation, use many fixed evaluation seeds, a verified success criterion, and report confidence intervals.
• Save the exact training configuration, package versions, and evaluation code with the project.

7. Conclusion

The reported reward results show a large improvement after extended PPO training. The project demonstrates environment setup, model training, checkpointing, evaluation, visualization, and an educational reward-design experiment. The results are promising, with the caveat that landing indicators need a stronger success definition before being reported as a true landing success rate.

Prepared from the user's reported local evaluation output. This report does not independently rerun the model.